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The key to dinoflagellate (*Noctiluca scintillans*) blooming and outcompeting diatoms in winter off Pakistan, northern Arabian Sea

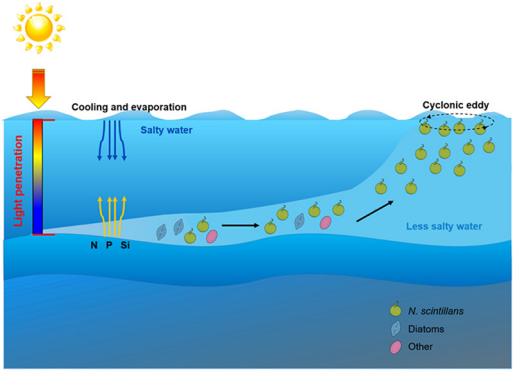
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HIGHLIGHTS

- Stratified water column and subsurface layer with less salty and ~24 °C are crucial for *N. scintillans* accumulation.
- Pumping effect of cyclonic eddy promotes *N. scintillans* upwelling from subsurface and blooming in surface water.
- Silicate limitation and organic nutrient accumulation promote *N. scintillans* outcompeting diatoms and bloom species turnover.

GRAPHICAL ABSTRACT



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ABSTRACT

The Arabian Sea is prone to large-scale algal blooms during winter monsoon annually. However, it is unclear why dinoflagellate, especially *Noctiluca scintillans*, replaced diatoms as the main bloom species. Based on in situ, remote sensing and numerical-model data off Pakistan, we found a stratified water with less salty, suitable temperature (~24 °C) and low-light conditions at the subsurface, as well as the organic nutrient accumulation and silicate limitation, were crucial for the growth of *N. scintillans* and outcompeting diatoms. The superposition of cyclonic eddy promoted *N. scintillans* pumping to surface and forming large-scale bloom. Subsequently, the shading effect of surface bloom caused the disappearance of subsurface chlorophyll maximum layer. This result suggests that the combined effects of nutrient structure and hydrodynamics play an important role in the prevalence of *N. scintillans*.

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1. Introduction

The Arabian Sea is one of the world's most productive sea areas, and it experiences large-scale seasonal blooms (Madhupratap et al., 1996; Prakash et al., 2008). Monsoon-driven physical processes characterize the area, including coastal upwelling under the southwest summer monsoon and convective mixing under the northeast winter monsoon (Ahmed et al., 2016; Centurioni et al., 2017; Morrison et al., 1999). These processes inject nutrients from subsurface into the surface layer to promote phytoplankton growth (Kumar and Prasad, 1996; Morrison et al., 1999). Atmospheric deposition is another important source of external nutrients because the Arabian Sea is surrounded by dry and dusty regions, and the dust from the land contains many nutrients and trace metals (Nezlin et al., 2010; Pease et al., 1998). Chlorophyll concentration is found closely related with precipitation and aerosol data (Nezlin et al., 2010). In addition, blooms of the dinoflagellate, such as *Noctiluca scintillans* are considered to be an outcome of eutrophication. Sewage from cities and agriculture enters the ocean and causes harmful algal blooms (Al Shehhi et al., 2014; Cozzi et al., 2014; Glibert et al., 2002; Kumar et al., 2010).

Bloom species succession from diatoms to dinoflagellates in the Arabian Sea has appeared since 2000s (Chaghtai and Saifullah, 2006; Gomes et al., 2014; Hansen et al., 2004). Although the enrichment of nutrients increases the possibility of algal blooms, there is no consensus about why diatoms are replaced by dinoflagellates and why *N. scintillans* become the dominant species. Diatom growth is limited by ambient nutrient supply, whereas dinoflagellates, especially *N. scintillans*, can rely on flagella-driven swimming and buoyancy of transparent cells to obtain nutrients actively. The waters in the Arabian Sea are stably stratified and light limited in winter (Kumar and Prasad, 1996; Rajaneesh et al., 2017). In such an environment, dinoflagellates have a survival advantage over diatoms (Horner et al., 1997). In addition to photosynthesis, phagocytizing organic particles also helps *N. scintillans* to obtain energy, and this process may also inhibit the growth of diatoms and other microalgae (Al-Azri et al., 2007; Goes et al., 2018; Hansen et al., 2004; Miyaguchi et al., 2006). On the other hand, apart from traditional nitrogen forms like nitrate, dinoflagellates have a strong affinity for organic forms like urea (Bronk et al., 2006; Glibert, 2017). Their growth is better when urea is added than when nitrate or ammonia is added (Tan et al., 2016). Therefore, the accumulation of organic nutrients caused by long-term and frequent blooms and the riverine discharge from the use of nitrogen fertilizers in agriculture promotes *N. scintillans* bloom (Glibert et al., 2016; Goes et al., 2018). Furthermore, the green type of *N. scintillans* adapts better to the oxygen deficient and light-limited conditions in the Arabian Sea because the endosymbiont, *Pedinomonas noctilucae*, increases its photosynthetic capacity by 25–300% (Gomes et al., 2014; Wang et al., 2016).

Although high primary productivity and seasonal blooms are frequently reported in the Arabian Sea (Ahmed et al., 2016; Krey and Babenerd, 1976), most of them are limited to the western coast of India or limited in the northern boundary of 22°N. The potential mechanism responsible for the succession of bloom species from diatom to *N. scintillans* is poorly understood. This study is based on the first China-Pakistan joint expedition to the northern Arabian Sea (22–24°N, 62–64°E). We aim to identify the causative factors leading to the bloom species succession and the mechanism underlying the formation of *N. scintillans* blooms in the Arabian Sea by using in situ hydrological (e.g., temperature and salinity), chemical (e.g., dissolved oxygen concentration and nutrient condition), and biological (e.g., phytoplankton biomass and composition) data. Also, by combing observed data with remote sensing and numerical-model data to understand the impact of dynamic hydrographic condition on the changes of the bloom.

2. Materials and methods

2.1. Study area and sampling

During the first China-Pakistan joint expedition to the northern Arabian Sea, the environmental and biological factors of the study area were investigated onboard research vessel R/V Shiyan 3 from January 19 to February 1, 2018. Fourteen stations along transect A (from S1 to S10) and B (from S11 to S14) (Fig. 1a). Seawater samples were taken using a Rosette sampler fitted with twelve 8 L Niskin bottles mounted on a Sea-Bird Electronics CTD (SBE-911 plus, Belleview, WA, USA) from seven depths (0.5, 25, 50, 75, 100, 150, and 200 m).

2.2. Data

Daily sea surface Chl *a* (MODIS-Aqua Level-3, 4 km resolution) was obtained from the NASA Distributed Active Archive Center (http://oceancolor.gsfc.nasa.gov/). Daily Sea Level Anomaly (SLA) and geostrophic current in correspondence of the observed cyclonic eddy were acquired from AVISO satellite altimeter with 1/4° grid (http://www.aviso.altimetry.fr). Daily salinity, including sea surface and vertical profile, was obtained from the Global Ocean 1/12° Physics Analysis product (numerical-model) at Copernicus Marine Environment Monitoring Service (CMEMS, http://marine.copernicus.eu).

Data from CMEMS has been analyzed in some studies (Hu et al., 2019; Lguensat et al., 2018; Miladinova et al., 2018). And this data has been proven to be reliable as comparing with satellite data at the same period (Hu et al., 2019). Also in the present study, the vertical profile of the salinity from CMEMS is consistent with our in situ observations (Fig. 3a, e).

2.3. In situ hydrographic and biological parameters

Salinity and temperature were measured by the CTD during the cruise, and the data were quality controlled and binned over 1 m depth intervals. Dissolved oxygen (DO) concentration was measured by Winkler titration method (Carritt and Carpenter, 1966). Nitrate, nitrite, ammonia, silicate and soluble reactive phosphorus (SRP) were analyzed using a flow injection analyzer (QuichChem8500, Lachat Inc., Loveland, CO, USA) (Grasshoff et al., 1983). Total dissolved nitrogen (TDN) and total dissolved phosphorus (TDP) were measured by the flow injection analyzer after being oxidized with persulphate in an alkaline medium (Grasshoff et al., 1983). Dissolved organic nitrogen (DON) content was calculated as the difference between TDN and dissolved inorganic nitrogen (DIN, sum of nitrate, nitrite, and ammonia). Dissolved organic phosphorus (DOP) was calculated as the difference between TDP and SRP. The profiles of dissolved nutrients within upper 100 m were presented because the Chl *a* mainly distributed within this layer.

Seawater was sequentially filtered through 20 μm pore-size nylon-net filter, 3 μm polycarbonate filter (Millipore) and 0.7 μm glass fiber filter (25 mm, Whatman GF/F) successively to measure micro-, nano- and pico-size Chl *a*, respectively. Total Chl *a* is the sum of micro-, nano- and pico-Chl *a* concentration. Filters were wrapped in aluminum foil and stored in liquid nitrogen, then measured by Turner Designs fluorometer 10-AU following the method of Parsons et al. (1984).

2.4. Phytoplankton identification and biomass calculation

For microscopic species composition, we collected 2 L seawater and fixed with 30 mL Lugol's solution (Parsons et al., 1984). We concentrated the fixed samples to 40 mL by siphoning after 24 h of precipitation. One milliliter condensed sample for species identification and numeration using an inverted microscope at ×200 or ×400 magnifications (CX21, OLYMPUS, Japan) (Utermöhl, 1958).

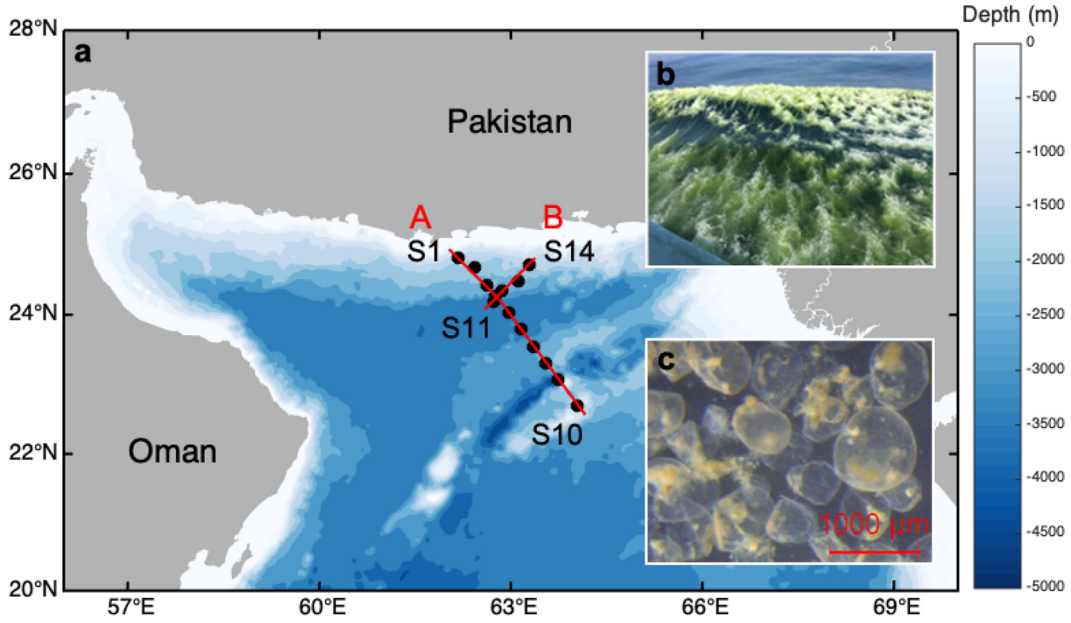


Fig. 1. (a) Map showing location of sampling points in the northern Arabian Sea (A: transect A, from S1 to S10; B: transect B, from S11 to S14); (b) real time shot of the surface *N. scintillans* bloom at S9 at 28 January 2018; (c) Single cells of *N. scintillans* collected at S9.

N. scintillans is larger than most other algal species with cell diameter reaches to 750 μm. Therefore, using cell biomass can more accurately reflect the phytoplankton community structure and energy distribution in marine ecosystems during the *N. scintillans* bloom (Hillebrand et al., 1999; Smayda, 1978). In this study, we obtained the biovolume of each algal species according to Napiórkowska-Krzebietke and Kobos (2016); and then, we multiplied the biovolume (μm^3) to cell abundance (cells L^{-1}) and single cell density (1 g cm^{-3}) to get the biomass.

2.5. Statistical analysis

Mixed layer (ML) depths were estimated as $\Delta\sigma_\theta = 0.03$ kg m^{-3} according to de Boyer Montégut et al. (2004). Surfer 10 software (Golden Software, Inc., Golden, CO, USA) was used to draw the figures. Redundancy analysis (RDA) was performed using CANOCO 5.0 software to explore how the surface phytoplankton community structure changed between non-bloom and bloom samples, as well as the impacts of environmental variables upon the bloom of *N. scintillans*. Because the *N. scintillans* bloom was concentrated in surface and Chl *a*, especially micro-size Chl *a* decreased dramatically from surface to deeper layer, the relative abundance of surface dominant algal groups were used for the analysis after being centered and standardized. All environmental variables were standardized.

3. Results

3.1. Time-series changes of remoting Chl a, sea level anomaly and salinity

The bloom outbroke during 25 Jan to 1 Feb (Fig. 2b) and this period is exactly the period of our onboard research and the high Chl *a* value was centered around the location of the real time shot of the surface *N. scintillans* bloom. The high Chl *a* region was corresponded to the cyclonic eddy which was indicated by a negative SLA (Fig. 2d). Surface salinity (SSS) from CMEMS was generally high along the investigated transect A (Fig. 2g–i).

3.2. Vertical profiles of temperature, salinity and dissolved oxygen

The water column was stratified. Salinity and temperature were high in the surface and became lower around 100 m depth (Fig. 3a, b). A significant convexity of the ML and an upwelling of less salty water from ~100 m depth to the surface layer were observed around S9 (Fig. 3a). Moreover, the upwelling process became stronger and more focused at S9. Also, the depth of ML became shallower from 17 Jan to 1 Feb (Fig. 3d–f) and the shallowest in bloom period (Fig. 3e). Coincidentally, the center of cyclonic eddy was located close to S9 in the bloom period (Fig. 2h). DO content was sufficient in the upper 100 m depth (6.67 ± 1.43 mg L^{-1}) but was <2.5 mg L^{-1} below (Fig. 3c).

3.3. Vertical profiles of nutrients and phytoplankton

Phytoplankton biomass (indicated by total Chl *a*) was mainly assembled at surface (0.56–1.91 μg L^{-1}) and dropped to 0.10 μg L^{-1} at 100 m depth (Fig. 4a). The highest surface biomass occurred at S9 (Fig. 4a) and consisted mostly of micro-sized cells (80.92%), which corresponded to the highest density of *N. scintillans* (Fig. 5). Dissolved organic nutrients accounted for a non-negligible proportion of total dissolved nutrients. The ratio of DOP to TDP (DOP%) and DON to TDN (DON%) were up to 79.1% and 43.15%, respectively (Fig. 4b, c). The DIN:DSi (dissolved silicon) ratio was 2:1 which implied a limitation of DSi (Fig. 4d). Additionally, dense sinking organic debris at S9 weaken the transmission of light in upper layer (Video in Supporting information).

A total number of 25 phytoplankton genera comprised of 16 diatoms, 8 dinoflagellates and one Chrysophyta were identified in surface water during the study period (Table 1). Total phytoplankton biomass was dominated by diatoms within non-bloom stations (47.17%, Fig. 5a). However, *N. scintillans* was overwhelmingly dominated within bloom stations (Fig. 5a). RDA showed that total Chl *a* was elevated within bloom area (Fig. 5b). Micro-sized Chl *a*, which was closely related to *N. scintillans* became dominant, only few chain-formed or elongated diatoms coexisted with abundant *N. scintillans*, such as *Nitzschi* spp., *Skeletonema costatum* and *Thalassiothrix* sp. High DIN:DSi ratio and DON were also positively correlated to the bloom. Total Chl *a* was lower within non-bloom stations and was mainly supported by nano- and pico-fraction. But species was more abundant and contained more diatoms and small dinoflagellates (Fig. 5b).

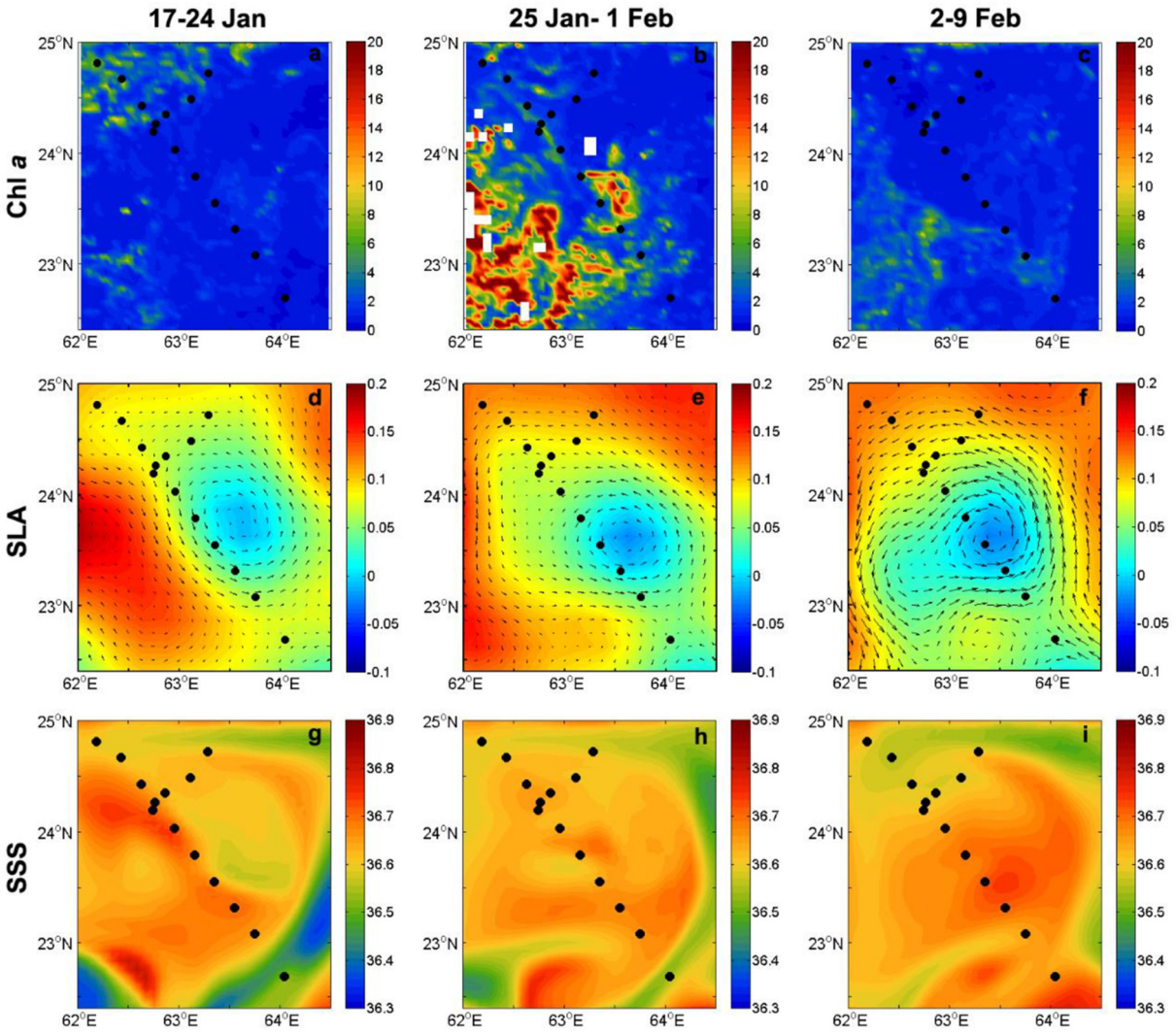


Fig. 2. Time series of eight-day average composition images of (a, b, c) sea surface MODIS-Aqua Chl *a* (μg L^{-1}); (d, e, f) sea level anomaly (SLA, m) and sea surface geostrophic current; (g, h, i) sea surface salinity from CMEMS (black dots present sample stations, the locations are the same as Fig. 1).

4. Discussion

4.1. Promotion of N. scintillans bloom by convection and cyclonic eddy

In this study, observed data showed a high surface salinity (Fig. 3a), and both remote sensing data and numerical-model data showed the salty surface water preceded the presence of high Chl *a* condition (Figs. 2g–i, 3d–f). This pattern demonstrates that convective mixing caused by surface cooling-driven excessive evaporation was the main cause of high productivity in winter (Al Shehhi et al., 2014; Garrison et al., 2000; Gomes et al., 2008; Kumar and Prasad, 1996; Prakash et al., 2008). Convection caused nutrient injection into the surface from the bottom of the ML (Madhupratap et al., 1996). Additionally, we observed a cyclonic eddy and the maximum surface Chl *a* occurred at the center of the eddy. Meanwhile, the profiles of temperature and salinity showed an obvious convexity of the ML and upwelling of less salty water from subsurface to surface at the center of the eddy (Fig. 3a, d–f). These evidences indicate the importance of the cyclonic eddy for upwelling phytoplankton cells and forming bloom in the surface. Under the effect of the cyclonic eddy, Chl *a* content was more than eight times higher and nitrate increased to 5 μmol L^{-1} (Naik et al., 2008). Chl *a* reached 1.9 μg L^{-1} in our study, mainly contributed by *N. scintillans*, which was consistent with previous reports (Lotliker et al., 2018; Naik et al., 2008; Piontkovski and Claereboudt, 2012). Furthermore, studies have shown that this process can be further enhanced when cyclone and convection are superimposed (Labiosa and Arrigo, 2003).

4.2. Stratification of water column initiates N. scintillans bloom

The water column was strongly stratified during the survey: the seawater salinity was higher in the surface (36.6 ± 0.13) than in the subsurface (~100 m depth) (36.3 ± 0.17) (Fig. 3a). The hydrological characteristics of the stratified water column with weak light radiation made *N. scintillans* more competitive than diatoms, because its swimming ability and the buoyancy of transparent cell allowed it to obtain nutrition and food actively in a stable water column with low turbulence (Miyaguchi et al., 2006). For diatoms, they only obtain food by absorbing nutrients from turbulent water. Additionally, the green type of *N. scintillans* that inhabits the northern Arabian Sea contained endosymbiotic *P. noctilucae*, which increases its photosynthetic capacity by 25–300% (Gomes et al., 2014; Wang et al., 2016).

In this study, the area of the cyclonic eddy coincided with the location of the surging of less salty subsurface water and corresponded to the maximum surface Chl *a* (S9, Fig. 3a). The regression analysis showed a significant negative correlation between the micro-Chl *a* concentration and salinity ($r = -0.88$, $n = 42$, $p < 0.01$), and the surface salinity at station S9 was 36.1, which was close to the range of salinity suitable

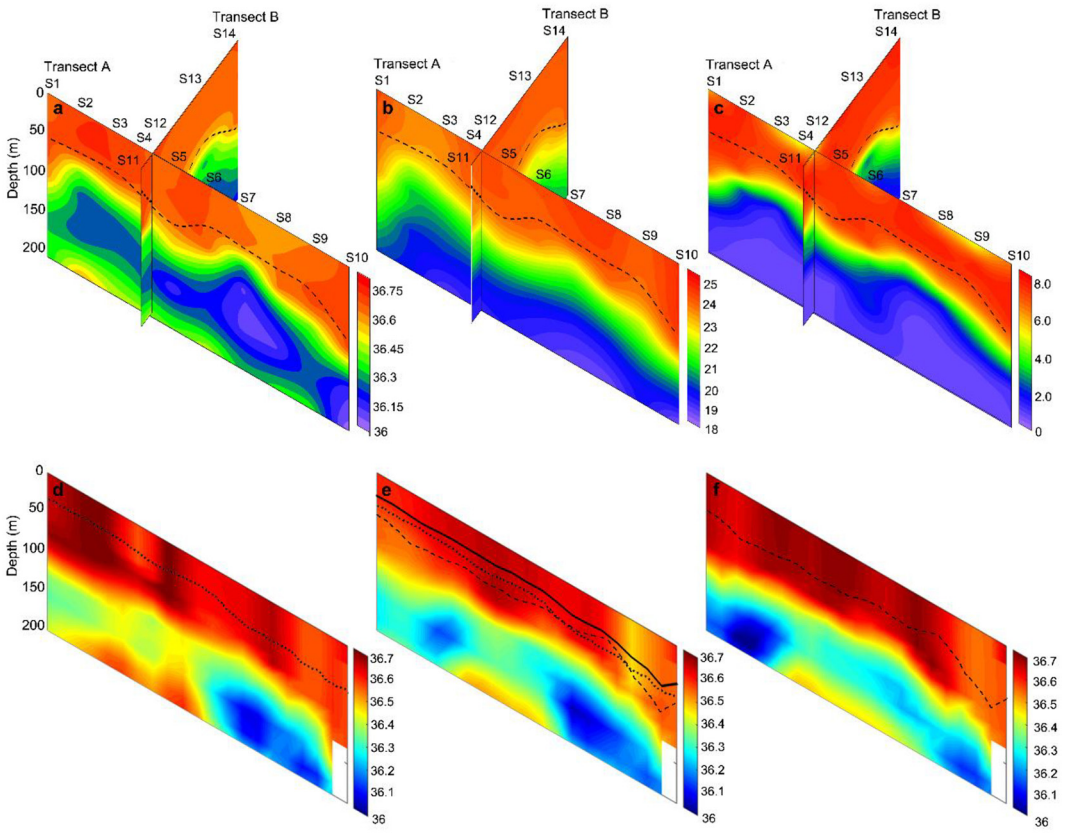


Fig. 3. Profiles of in-situ (a) salinity, (b) temperature (°C) and (c) DO (mg L^{-1}) with mixed layer depth indicated by dashed line; temporal eight-days average profiles of salinity with mixed layer depth derived from CMEMS for (d) 17–24 Jan, (e) 25 Jan-1 Feb and (f) 2–9 Feb; in (e), the dot point and dash line is the mixed layer depth of pre- (d) and after-bloom (f), respectively.

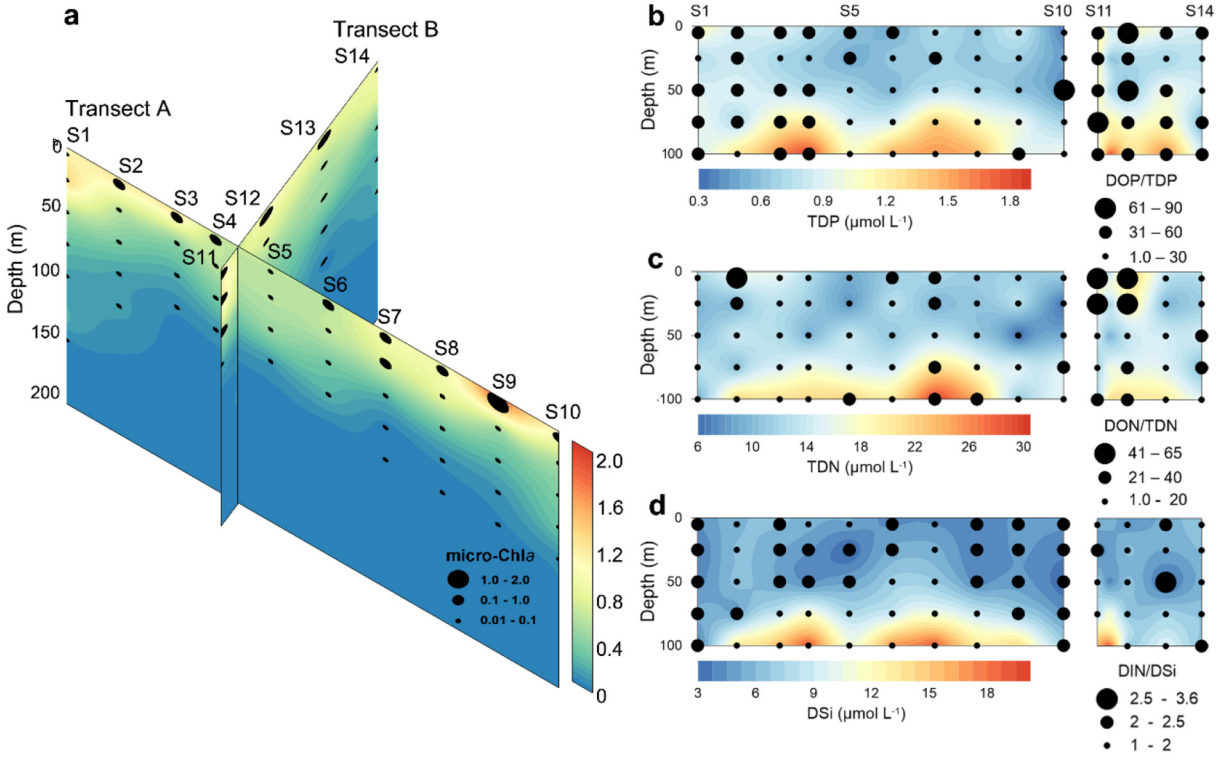


Fig. 4. Vertical distributions of (a) total Chl *a* (μg L^{-1}) (contour map) and micro-sized Chl *a* (dot) concentrations; (b) TDP (μmol L^{-1}) (contour map) and proportion of DOP to TDP (%) (dot); (c) TDN (μmol L^{-1}) (contour map) and proportion of DON to TDN (%) (dot); (d) DSi (μmol L^{-1}) (contour map) and molar ratio of DIN to DSi (DIN/DSi, dot).

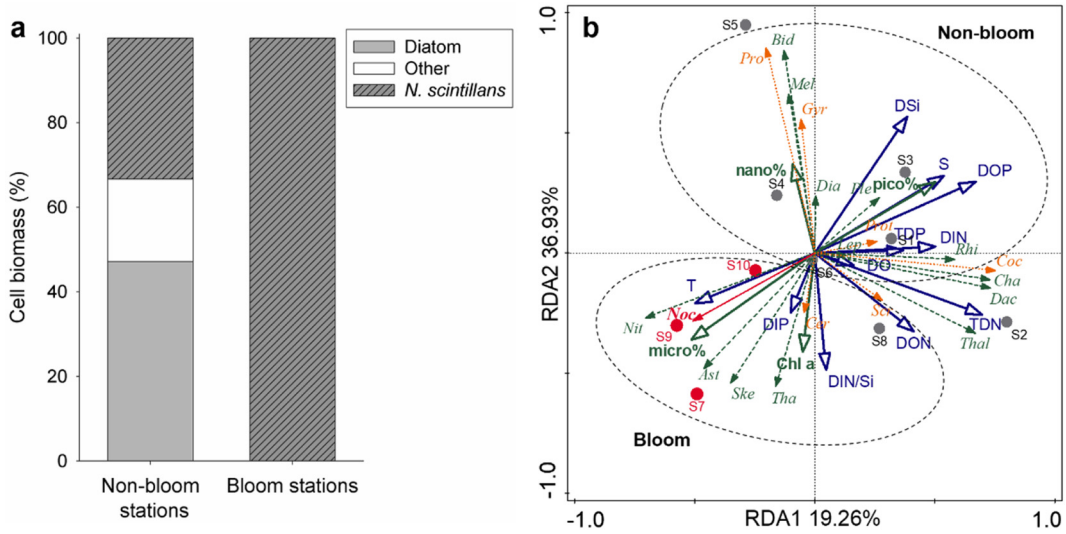


Fig. 5. (a) Average biomass ratio of *Noctiluca scintillans* within representative non-bloom (47.17%) and bloom stations (ca. 100%) and (b) correlation triplot of phytoplankton community structure and its relationship with environmental variables based on redundancy analysis (RDA). Constrained axes RDA1 and RDA2 account for 19.26% and 36.93% of total variations, respectively. Chl *a*, total Chl *a* concentration (μg L^{-1}); micro%, proportion of micro-size Chl *a* to total Chl *a* (same as nano% and pico%); Ast, *Asterionella*.; Bid, *Biddulphia*; Cha, *Chaetoceros*; Dac, *Dactyliosolen*; Dia, *Diatoma*; Lep, *Leptocylindrus*; Mel, *Melosira*; Nit, *Nitzschia*; Ple, *Pleurosigma*; Rhi, *Rhizosolenia*; Ske, *Skeletonema*; Thal, *Thalassiosira*; Coc, *Cochlodinium*; Cer, *Ceralium*; Gyr, *Gyrodinium*; Noc, *Noctiluca scintillans*; Pro, *Prorocentrum*; Prot, *Protoperidinium*; Scr, *Scripsiella trochoidea*.

for *N. scintillans* growth (Huang and Qi, 1997). The surface temperature (24.03 ± 0.32 °C) also was optimal for *N. scintillans* growth (25 °C) (Mohamed and Mesaad, 2007; Tsai et al., 2018; Turkoglu, 2013) and allowed *N. scintillans* to increase its growth rate 8-fold in the light-sufficient surface layer (Gomes et al., 2018). Results of this study showed that the initial cell accumulation of *N. scintillans* occurred in the stratified water column in the low salt subsurface layer. Cells were brought to the surface layer via the upward transport of the cyclone, forming a large-scale surface bloom under suitable temperature and light conditions. Piontkovski et al. (2017) reported a similar phenomenon in the Arabian Sea.

Table 1
Averaged biomass (μg L^{-1}) of each phytoplankton species of representative non-bloom (i.e., S3, S4 and S5) and bloom stations (i.e., S7, S8 and S9) in surface water of the northern Arabian Sea (mean ± SD, $n = 3$).

Species	Non-bloom stations	Bloom-stations
Diatoms	127.74 ± 139.06	134.82 ± 24.57
Asterionella japonica	–	2.84 ± 4.93
Biddulphia mobiliensis	5 ± 4.33	–
Chaetoceros spp.	0.56 ± 0.97	0.56 ± 0.97
Coscinodiscus sp.	15.7 ± 27.19	–
Dactyliosolen fragilissimus	0.45 ± 0.78	–
Diatoma sp.	0.52 ± 0.45	–
Leptocylindrus danicus	0.04 ± 0.07	0.08 ± 0.14
Melosira sp.	40.82 ± 36.92	39.25 ± 39.25
Nitzschia spp.	2.83 ± 0.57	10.78 ± 6.89
Planktoniella formosa	15.7 ± 27.19	–
Pleurosigma directum	14.06 ± 24.36	–
Pseudo-nitzschia pungens	–	0.25 ± 0.43
Rhizosolenia spp.	31.92 ± 49.95	78.89 ± 46.66
Skeletonema costatum	–	1.78 ± 2.56
Thalassionema sp.	0.13 ± 0.23	0.39 ± 0.39
Thalassiosira sp.	–	–
Dinoflagellates	$2.95 \times 10^4 \pm 5.10 \times 10^4$	$1.47 \times 10^5 \pm 6.74 \times 10^4$
Ceralium sp.	–	10.90 ± 9.44
Cochlodinium spp.	66.99 ± 66.99	22.33 ± 38.67
Dinophysis sp.	–	–
Gyrodinium sp.	1.4 ± 1.21	2.09 ± 2.09
Noctiluca scintillans	$2.94 \times 10^4 \pm 5.10 \times 10^4$	$1.47 \times 10^5 \pm 6.75 \times 10^4$
Prorocentrum sp.	24.42 ± 15.99	13.96 ± 12.09
Protoperidinium sp.	–	–
Scrpsiella trochoidea	0.28 ± 0.48	1.40 ± 2.42
Chrysophyta	–	3.49 ± 6.04
Dictyocha fibula	–	3.49 ± 6.04
Total biomass	$2.96 \times 10^4 \pm 5.09 \times 10^4$	$1.47 \times 10^5 \pm 6.74 \times 10^4$

4.3. Organic nutrient accumulation and silicate limitation drive phytoplankton succession

Organic nutrients accounted for large part of total dissolved nutrients in the surveyed area, especially in the nearshore stations (Fig. 4b, c). It is consistent with the results of Gomes et al. (2018) and Harrison et al. (2011) that indicated the green *N. scintillans* often outbreaks in the coastal agricultural economic zone of Southeast Asian countries, where urea is widely used. Moreover, the *N. scintillans* grows lager and contains more endosymbiotic algae in urea-rich seawater, compared with nitrate- or ammonia-rich seawater (Tan et al., 2016). Additionally, most dinoflagellates have been observed to secrete phosphodiesterase to decompose organic phosphorus into inorganic phosphorus (Rivkin and Swift, 1980; Sakshaug et al., 1984).

Silicon limitation is another major factor that contribute to the succession of dominant bloom species. Generally, the DIN:DSi ratio is 1:1 in the oceans, while this ratio was >1 in the northwestern Arabian Sea from 2003 to 2016 (Prakash et al., 2017), in consistent with our results (DIN:DSi, ~2:1) (Fig. 4d). Moreover, the enhanced density stratification inhibited the transport of silicate from deeper water to the upper layer. Consequently, diatoms, which require silicate, are restricted and *N. scintillans* become dominated (Lotliker et al., 2018; Prakash et al., 2017; Wasmund et al., 2013). On the other hand, *N. scintillans* inhibits the growth of other microalgae through phagocytosis (Al-Azri et al., 2007; Gomes et al., 2018). Therefore, waters with rich organic matters provide a food source for *N. scintillans* in the form of nutrients and heterotrophic nutrition, helped *N. scintillans* maintain strong photosynthesis in the light- and oxygen-deficient waters (Sweeney, 1976; Wang et al., 2016).

4.4. Disappearance of subsurface Chl *a* maximum layer driven by *N. scintillans* bloom

Chl *a* concentration was highest in the surface layer (Fig. 4a) and formed a thick green barrier (Fig. 1b) because of the upward transport of cells by convection and the cyclonic eddy, also because of the positive buoyancy of the transparent cells. Ultimately, the surface bloom blocked the transmission of sunlight, and simultaneously produced a large

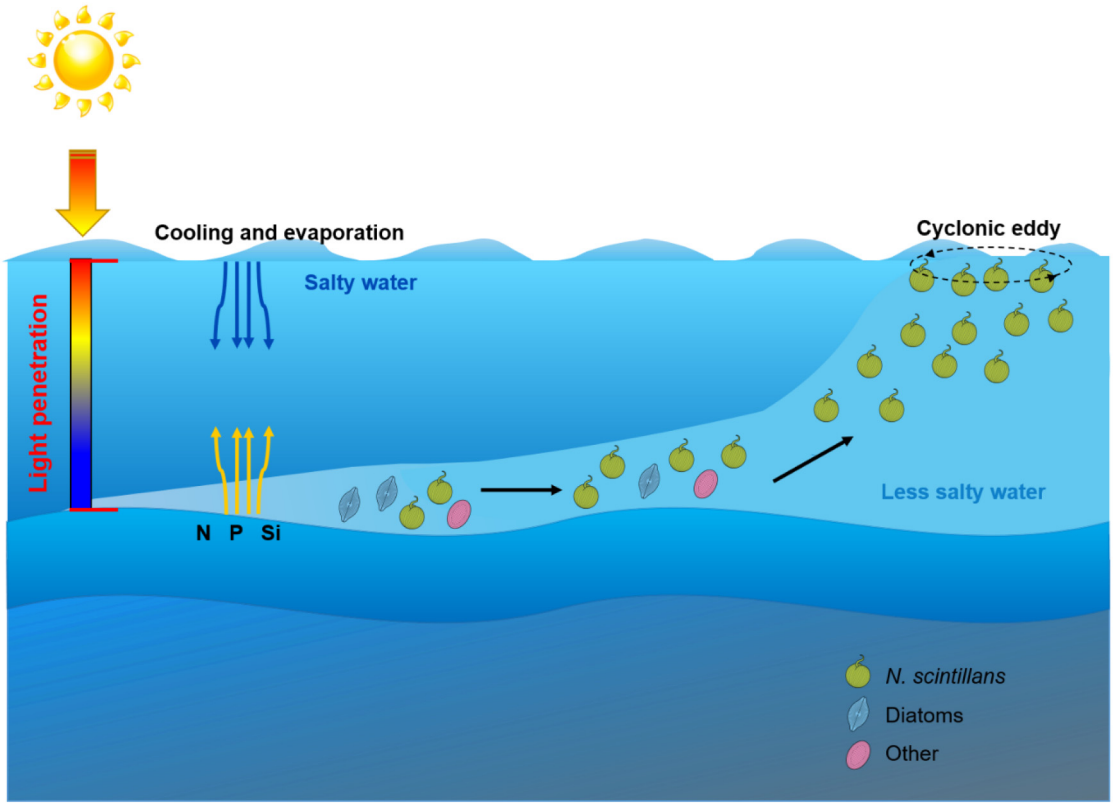


Fig. 6. Conceptual pattern of the formation process of *N. scintillans* bloom.

amount of sinking organic debris (Video in Supporting information), which further reduced water transparency by absorbing and dispersing sunlight. *N. scintillans* growth was also fueled by feeding on extraneous phytoplankton from the subsurface layer. Several studies recently demonstrated that mixotrophic green *N. scintillans* cells require both photoautotrophic and heterotrophic processes to occur simultaneously to achieve maximum cell growth rates, even under sufficient irradiance (Gomes et al., 2018). Therefore, feeding on other phytoplankton species in the subsurface water layer was enhanced during the formation stage of the *N. scintillans* bloom, which reduced the subsurface Chl *a* concentration (Bhattathiri et al., 1996; Lotliker et al., 2018; Madhupratap et al., 1996). In summary, reduction of solar irradiance and predation on other phytoplankton species by the blooming *N. scintillans* were the main reasons for the disappearance of the subsurface Chl *a* maximum layer.

5. Conclusions

We conclude that stable stratification and subsurface lower salinity water caused by surface cooling and excessive evaporation, contributed to the accumulation of *N. scintillans* cells in the study area. Additionally, high DIN:Si, together with long-term organic nutrient accumulation due to diatom blooms and terrestrial input of rich organic nutrients in the euphotic zone, provided the favorable conditions for *N. scintillans* to outcompete diatoms. Thus, large-scale *N. scintillans* bloom in surface water occurred under the conditions of upward convection and the presence of a cyclonic eddy (Fig. 6).

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2019.07.202.

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